

GREEN MIND SYSTEM: An IoT-Driven Approach for Smart Waste Monitoring and Management

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DECLARATION

We hereby declare that this report entitled **GREEN MIND SYSTEM: An IoT-Driven Approach for Smart Waste Monitoring and Management** is our own original work carried out as a under graduate student Netaji Subhash Engineering College except to the extent that assistances from other sources are duly acknowledged.

All sources used for this project report have been fully and properly cited. It contains no material which to a substantial extent has been submitted for the award of any degree/diploma in any institute or has been published in any form, except where due acknowledgement is made.

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ABSTRACT

This project introduces an **intelligent waste management solution** that automates the critical early stages of disposal: **detection, segregation, and monitoring**. Using the **NodeMCU ESP32** as the central microcontroller, the system integrates multiple sensors and actuators to create a responsive, IoT-enabled smart bin. Two **IR sensors** are deployed—one to monitor the **fill level** of the bin and another to detect **waste entry**. A **proximity switch** ensures accurate detection of incoming waste, while a **rain/moisture sensor** identifies wet conditions. Mechanical operations are handled by a **stepper motor** for precise movement and a **servo motor** for directional control. An **M23 gas sensor** monitors gases emitted from decomposing wet waste, providing early alerts for hazardous conditions. Data collected is transmitted via Wi-Fi for cloud-based analytics. In future iterations, **AI and ML algorithms** will be integrated to enhance waste classification, predictive analysis, and intelligent decision-making.

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LIST OF COMPONENTS and EQUIPMENTS

SL NO.	NAME OF THE COMPONENT and EQUIPMENTS	SPECIFICATIONS
1	RAIN/SOIL SENSOR (MOISTURE)	—
2	IR SENSOR	—
3	PROXIMITY SWITCH	—
4	SERVO MOTOR	MG996R
5	STEPPER MOTOR	—
6	ESP32 MICROCONTROLLER	<i>NODE-MCU</i>

Chapter-1 INTRODUCTION

This chapter provides the introductory concept and motive of the proposed work. It presents aim of the project, problem statement and scope of the study.

1.1 AIM OF THE PROJECT

Urban waste management faces critical challenges in the 21st century, with increasing waste generation and inadequate segregation practices leading to environmental degradation. This project envisions an intelligent waste management solution that automates segregation at the source, utilizing IoT technologies for real-time monitoring and future AI capabilities for enhanced accuracy.

The smart dustbin system aims to:

- Automate waste classification into wet, dry, and metal categories.
- Enable remote monitoring and maintenance alerts.
- Detect hazardous conditions through gas sensing.
- Provide a foundation for AI-driven waste management systems.

Hardware & Sensor Architecture

To achieve these objectives, the system is built around a compact, IoT-enabled microcontroller platform like the NodeMCU ESP32, which serves as the central brain for sensor data acquisition and cloud communication. The automated sorting mechanism relies on a coordinated network of specific hardware components:

- **Detection & Classification:** Infrared (IR) sensors are positioned to instantly detect the presence of deposited waste. A combination of rain/moisture sensors and proximity switches then rapidly analyzes the material's properties to accurately classify it as wet organic matter, dry waste, or metal.
- **Automated Routing:** Once the waste is classified, precision stepper and servo motors actuate a mechanical sorting flap or tray, guiding the item directly into its designated internal compartment.
- **Safety Monitoring:** To ensure environmental safety and prevent combustion from decomposing waste, an MQ3 gas sensor is integrated to continuously monitor the internal bin environment for hazardous emissions or flammable vapors.

Software & Cloud Integration

The embedded components are programmed using C++ to ensure fast, low-latency processing for real-time sensor polling and motor control. The ESP32 node continuously transmits bin capacity levels, operational statuses, and gas anomaly alerts to a centralized cloud dashboard. This real-time visibility allows municipalities to dynamically optimize garbage collection routes, significantly reducing fleet fuel consumption and completely eliminating bin overflow incidents.

Future Scope: AI & Machine Learning

While the immediate architecture relies on deterministic sensor logic, the system acts as a crucial data-gathering node. By logging continuous disposal metrics, it builds a rich dataset necessary for training advanced artificial intelligence models. Utilizing Python-based machine learning algorithms, future iterations of the system could perform predictive maintenance, forecast neighborhood waste generation trends, and integrate computer vision to handle highly complex, multi-material sorting.

1.2 PROBLEM STATEMENT

Current waste management systems suffer from several critical limitations:

- **Manual Segregation Issues:** Traditional bins require users to manually segregate waste, leading to high contamination rates and reduced recycling efficiency. Studies indicate that improper segregation reduces recyclable material recovery by up to 40%.
- **Health and Safety Concerns:** Accumulated organic waste produces methane and other harmful gases, creating health hazards. Overflowing bins attract pests and spread diseases, particularly in densely populated areas.
- **Monitoring Gaps:** Existing bins lack real-time status monitoring, resulting in delayed maintenance, overflow situations, and inefficient collection routes. Municipal authorities often rely on fixed schedules rather than actual bin status.

- **Scalability Limitations:** Manual waste management systems cannot scale effectively with urban population growth and increasing waste generation rates.

1.3 SCOPE OF THE PROJECT

Here is two phase of the scope of this project:

Phase 1: Primary Goals & Hardware Implementation

- **Automated Waste Segregation Mechanism:** Implement a precise sorting architecture using a central microcontroller (such as a NodeMCU ESP32) to process inputs from a network of specialized sensors. By combining rain/moisture sensors to identify wet organic matter and proximity/inductive switches for metallic and dry waste, the system will actuate stepper and servo motors to physically route discarded items into their correct internal compartments in real time.
- **IoT-Enabled Real-Time Monitoring:** Establish seamless cloud connectivity by integrating the embedded system with the Blynk application. This creates a centralized, remote dashboard where administrators can continuously monitor the operational status, connectivity, and health of distributed bins.
- **Hazardous Condition Detection:** Deploy a dedicated gas sensor (such as an M23 or similar MQ-series module) within the bin's enclosure to constantly monitor for combustible methane buildups, toxic chemical emissions, or decomposing hazards, instantly flagging dangerous conditions before they escalate.
- **Capacity Tracking via IR Sensing:** Utilize infrared (IR) proximity sensors strategically placed at the top of each internal segregation compartment to continuously measure fill levels. This prevents unhygienic overflow and allows for dynamic, data-driven collection schedules.
- **Automated Maintenance & System Alerts:** Program the system to autonomously generate push notifications for critical events. The system will instantly alert administrators regarding sensor malfunctions, motor stalls, or maximum capacity thresholds, ensuring minimal downtime and efficient dispatch of maintenance crews.

Phase 2: Future Enhancements & AI Integration

- **Computer Vision via ESP32-CAM:** Upgrade the primary detection mechanism by incorporating an ESP32-CAM module. This transitions the system from purely physical property-based sensor sorting to the visual identification of complex or mixed-material waste items.
- **AI/ML Model Deployment:** Develop, train, and deploy custom Machine Learning algorithms to classify a much wider variety of waste materials (e.g., distinguishing between specific grades of recyclable plastics and non-recyclables) with increasing accuracy over time.
- **Non-Invasive Image Processing:** Augment the physical IR level sensors with advanced image-processing techniques. By utilizing the camera feed, the system can visually estimate the exact volume, density, and distribution of waste inside the bin without relying on physical contact sensors.
- **Advanced Data Analytics & Predictive Modeling:** Aggregate long-term IoT data to map neighborhood-level waste generation trends, seasonal peaks, and disposal habits. This analytical foundation will enable predictive modeling for municipal fleet routing, effectively reducing fuel consumption, operational costs, and the overall carbon footprint of waste management.

Chapter-2 LITERATURE REVIEW

This chapter reviews the related work and existing literature relevant to the proposed system. The development of smart dustbins has evolved significantly over the last decade, shifting from simple mechanical automation to complex IoT and AI-based systems. This review categorizes existing research into three key areas relevant to this project: Sensor-Based Segregation, IoT Monitoring, and Visual Artificial Intelligence.

2.1 SENSOR-BASED WASTE SEGREGATION

Early research in automated waste management predominantly focused on segregating materials based on their inherent physical and electrical properties [1],[2]. While foundational, these methods presented both mechanical and logical limitations.

- **2.1.1 Moisture and Inductive Sensing:** Studies have consistently established that combining an inductive proximity sensor (for detecting ferromagnetic and conductive metals) with a moisture/capacitive sensor (for detecting organic content) is the most cost-effective architecture for categorizing waste into Metal, Wet, and Dry bins. However, a study by Shaikh et al. (2022) highlighted a critical edge case: the "shielding effect." While reliable for exposed organic matter, this method frequently results in false negatives when wet waste is encased in dry plastic bags (polythene), leading to the misclassification of organics as dry waste [1].
- **2.1.2 Mechanism Design & Actuation:** Literature analyzing the physical sorting actuators suggests that "flap-based" or "rotating disc" mechanisms—typically driven by PWM-controlled servo motors—are the industry standard for low-cost prototypes. Nevertheless, Kumar (2021) identified significant mechanical reliability issues, noting that single-stage separation (processing all sensors simultaneously) frequently causes physical jamming. The study advocates for a multi-stage, sequential gating architecture (e.g., verifying metal first, physically routing it, and then testing for moisture), which drastically reduces mechanical stress and sorting errors [2].

2.2 IOT-ENABLED MONITORING AND SAFETY

With the advent of Industry 4.0, the paradigm of municipal solid waste (MSW) management has shifted from reactive models (scheduled collection or overflowing bins) to proactive, data-driven frameworks.

- **2.2.1 Real-Time Telemetry and Alerts:** Research by Anitha (2019) documented the transition from legacy GSM-based SMS alert systems—which were plagued by high operational costs and limited data bandwidth—to Wi-Fi-based IoT cloud architectures. Utilizing platforms like Blynk and ThingSpeak allows for bidirectional communication and continuous time-series data logging. This enables administrators to visualize fill-level trends over time rather than relying on binary, one-time overflow alerts [3].
- **2.2.2 Hazard Detection & Environmental Safety:** Monitoring the internal microclimate of a waste bin has become a mandatory safety standard in contemporary literature. Bajaj and Reddy (2018) emphasized that the anaerobic decomposition of trapped wet waste rapidly generates combustible Methane (CH_4) and toxic Ammonia (NH_3). Integrating specific gas sensors (such as the MQ series) is essential to preemptively flag fire hazards, mitigate explosive buildups, and control severe public odor nuisances—a critical layer of intelligence absent in rudimentary systems [4].

2.3 COMPUTER VISION AND AI

Recent publications [5]-[7] spanning from 2022 to 2025 demonstrate a distinct pivot toward optical and acoustic machine learning, addressing the inherent limitations of "blind" contact sensors.

- **2.3.1 Visual Classification and CNNs:** A pivotal study by Sheng et al. (2022) utilizing the ESP32-CAM module introduced the framework of "Visual Waste Verification." The authors posited that utilizing Convolutional Neural Networks (CNNs) allows the system to identify waste via spatial features (shape, texture, and color)

without requiring physical contact. This optical approach effectively resolves the "wet waste in a plastic bag" dilemma by recognizing the plastic bag itself and routing it accordingly [5].

- **2.3.2 "ConvoWaste" (Deep Learning Segregation):** Nafiz et al. (2023) significantly advanced this field with "ConvoWaste," an automated segregation architecture driven by a Deep Convolutional Neural Network (DCNN). Their model successfully classifies waste into six distinct granular categories with an impressive 98% accuracy rate. Furthermore, the system establishes a holistic ecosystem by seamlessly merging computer vision inference with servo-motor actuation for physical sorting, alongside traditional IoT sensors for capacity monitoring, resulting in a highly scalable and cost-effective urban solution [6].
- **2.3.3 Smart Sort (Acoustic Edge AI):** In a novel departure from purely optical AI, the paper "Smart Sort: AI-Powered Waste Segregation Using Edge Impulse Studio" introduces acoustic-based classification. The authors utilize Edge Impulse to train machine learning models directly on microcontroller edge devices. Instead of capturing images, the system analyzes the unique audio signals (impact acoustics and material resonance) generated when an item strikes the bin's surface. By extracting audio features like spectrograms, the system can accurately classify materials like glass, metal, or plastic, offering a highly efficient, privacy-compliant, and low-power alternative to camera-based networks [7].

Chapter-3 FRAMEWORK and METHODOLOGY

This section outlines the conceptual framework and details the methodological approach of the proposed system.

3. SYSTEM ARCHITECTURE OVERVIEW

The proposed Smart Dustbin System is designed using a modular and scalable architecture that integrates sensing, processing, actuation, wireless communication, and cloud-based monitoring. The architecture is centred around the **ESP32 NodeMCU**, which functions as the main processing unit and coordinates the operation of all hardware and software components. The system automatically identifies different categories of waste using multiple sensors, performs intelligent waste segregation through servo-controlled mechanisms, monitors the fill level of each waste compartment, detects the presence of harmful gases generated from decomposing waste, and transmits real-time data to the cloud using Wi-Fi technology.

The modular design ensures that each subsystem operates independently while maintaining seamless communication with the central controller. This architecture simplifies troubleshooting, enables future hardware upgrades, and improves the overall reliability and efficiency of the system.

The complete system architecture consists of the following functional layers:

Input Layer

The input layer is responsible for collecting information from the surrounding environment and the waste materials deposited into the dustbin. Various sensors are integrated to perform specific detection tasks.

- Moisture Sensor for identifying wet and dry waste.
- Inductive Proximity Sensor for detecting metallic objects.
- IR Proximity Sensors for monitoring waste levels in different compartments.
- Gas Sensor for monitoring harmful gases generated from organic waste decomposition.

Processing Layer

The processing layer consists of the ESP32 NodeMCU microcontroller. It continuously receives signals from all sensors, processes the acquired data according to the embedded control algorithm, classifies the waste type, controls the movement of servo motors, and transmits real-time data to the IoT platform. The ESP32 also performs decision-making operations based on predefined threshold values and programmed logic.

Actuation Layer

The actuation layer includes servo motors that operate mechanical flaps inside the dustbin. Depending on the detected waste type, the ESP32 positions the corresponding flap to direct the waste into the appropriate compartment. This mechanism enables automatic segregation without any human intervention.

Communication Layer

The communication layer utilizes the inbuilt Wi-Fi capability of the ESP32 to establish an internet connection with the Blynk Cloud Platform. Sensor readings, compartment fill levels, and gas concentration data are uploaded continuously, enabling remote monitoring and instant notifications.

User Interface Layer

The user interface is developed using the Blynk Mobile Application. The application provides real-time visualization of sensor data, displays compartment status, sends alerts when bins become full or when hazardous gases exceed permissible limits, and allows users to monitor the system remotely from any location.

3.1 HARDWARE COMPONENTS

The Smart Dustbin System consists of several hardware modules that work together to perform automatic waste segregation and monitoring. Each hardware component has been carefully selected based on cost-effectiveness, reliability, compatibility, and ease of implementation.

3.1.1 ESP32 NodeMCU Controller

The **ESP32 NodeMCU** serves as the central processing unit of the entire system. It is a low-cost, high-performance microcontroller equipped with integrated Wi-Fi and Bluetooth connectivity, making it highly suitable for Internet of Things (IoT) applications.

The ESP32 continuously acquires sensor data, processes classification algorithms, controls servo motor movements, and uploads monitoring information to the cloud platform. Due to its powerful processor and wireless communication capability, it eliminates the need for additional communication modules, thereby reducing system complexity and cost.

Specifications

- Dual-Core Tensilica LX6 Processor operating up to 240 MHz
- 520 KB SRAM
- Integrated 2.4 GHz Wi-Fi
- Bluetooth Classic and Bluetooth Low Energy (BLE)
- 34 programmable GPIO pins
- Multiple ADC and DAC channels
- UART, SPI, I2C, PWM communication interfaces
- Operating Voltage: 3.3V

Functions

- Collects sensor data from all input devices.
- Processes waste classification algorithms.
- Controls servo motor positioning.
- Monitors compartment fill levels.
- Detects harmful gas concentration.
- Uploads sensor information to the Blynk Cloud.
- Sends alerts through the mobile application.

Advantages

- Integrated wireless communication.
- Low power consumption.
- Compact design.
- High processing speed.
- Large number of input/output pins.
- Excellent compatibility with Arduino IDE.

3.1.2 Sensing Subsystem

The sensing subsystem performs real-time monitoring of waste characteristics and environmental conditions. Multiple sensors operate simultaneously to improve the accuracy of waste segregation.

A. Capacitive Moisture Sensor

The capacitive moisture sensor is used to distinguish between wet and dry waste by measuring the moisture content present in the waste material.

Unlike resistive moisture sensors, capacitive sensors are resistant to corrosion and provide more stable and reliable readings over long-term operation.

Working Principle

The sensor measures changes in capacitance caused by varying moisture levels. Higher moisture content increases dielectric permittivity, resulting in changes in the analog output voltage. The ESP32 compares the measured value with predefined threshold values to determine whether the waste is wet or dry.

Specifications

- Operating Voltage: 3.3V–5V
- Output Type: Analog
- Low power consumption
- Corrosion resistant
- Fast response time

Applications

- Wet waste detection
- Organic waste identification
- Moisture monitoring

B. Inductive Proximity Sensor

The inductive proximity sensor is used to identify metallic waste such as iron, steel, nickel, and other ferromagnetic materials.

Working Principle

The sensor generates an electromagnetic field around its sensing face. When a metallic object enters this field, eddy currents are induced inside the object, causing changes in the oscillation amplitude. The sensor detects these changes and generates a digital HIGH output indicating the presence of metal.

Specifications

- Operating Voltage: 6–36V (depending on model)
- Digital Output
- Detection Distance: 0–15 mm
- Non-contact sensing
- Long operational life

Applications

- Metal waste segregation
- Ferrous object detection
- Industrial automation

C. IR Proximity Sensors

Infrared proximity sensors continuously monitor the fill level of each compartment inside the dustbin.

Working Principle

The sensor emits infrared light and measures the reflected signal. As the waste level approaches the sensor, the reflected intensity increases. Once the threshold distance is reached, the sensor outputs a digital signal indicating that the compartment is nearly full.

Specifications

- Operating Voltage: 3.3–5V
- Digital Output
- Adjustable sensing distance
- Fast response

Applications

- Bin level monitoring
- Overflow prevention
- Smart waste management

D. Gas Sensor (MQ-135)

The MQ-135 gas sensor detects harmful gases produced due to decomposition of organic waste. Monitoring gas concentration helps maintain hygiene and provides early warning of unpleasant odors and hazardous conditions.

Detectable Gases

- Ammonia (NH_3)
- Methane (CH_4)
- Hydrogen Sulfide (H_2S)
- Carbon Dioxide (CO_2)
- Benzene
- Smoke

Working Principle

The sensing element changes its electrical resistance according to the concentration of surrounding gases. The analog output voltage varies proportionally with gas concentration. The ESP32 continuously reads these values and compares them against predefined threshold limits.

Specifications

- Operating Voltage: 5V
- Analog Output
- High sensitivity
- Wide gas detection range
- Warm-up Time: 24–48 hours

Applications

- Air quality monitoring
- Odor detection
- Safety monitoring
- Smart waste management

3.1.3 Actuation Subsystem

The actuation subsystem performs the physical movement required for waste segregation.

Servo Motors (SG90 / MG996R)

Servo motors are employed to rotate internal flaps that guide waste into the appropriate compartment based on sensor classification.

Each servo receives PWM signals from the ESP32, allowing precise angular positioning between 0° and 180°.

Specifications

- Operating Voltage: 4.8V–6V
- Rotation Angle: 0°–180°
- Current Consumption: 100–500 mA
- PWM Controlled
- High positioning accuracy

Functions

- Opens segregation flap.
- Directs waste toward correct compartment.
- Returns flap to default position after operation.
- Enables automated sorting without manual intervention.

Power Supply Unit

The power supply subsystem provides stable electrical power to all electronic components.

A regulated **5V DC power supply** is used to operate the ESP32, sensors, and communication circuitry. Since servo motors require higher current during operation, a separate regulated **6V supply** may be used to prevent voltage fluctuations and improve system stability.

For portable installations, rechargeable lithium-ion batteries or UPS backup can also be incorporated to maintain uninterrupted operation during power failures.

Power Requirements

- ESP32: 3.3V (internally regulated)
 - Sensors: 3.3–5V
 - Servo Motors: 4.8–6V
 - Total Estimated Current: 1–2 A
-

3.2 SOFTWARE ARCHITECTURE

The software architecture provides intelligent control of the Smart Dustbin System. It integrates sensor acquisition, decision-making algorithms, servo motor control, cloud communication, and mobile monitoring into a unified software framework.

The software follows an event-driven architecture where sensor data are continuously monitored, processed, and transmitted to the cloud while simultaneously controlling hardware components.

3.2.1 Development Environment

The software is developed using the **Arduino IDE**, which provides a user-friendly integrated development environment for programming ESP32 microcontrollers.

Development Platform

- Arduino IDE
- ESP32 Board Package
- USB Serial Programming

Programming Language

- Embedded C
- C++
- Arduino Framework

Libraries Used

WiFi Library

- Connects ESP32 to wireless networks.
- Enables internet connectivity.
- Supports cloud communication.

Blynk Library

- Connects the ESP32 to the Blynk IoT platform.
- Uploads sensor readings in real time.
- Sends notifications and alerts.
- Displays live dashboard information.

Servo Library

- Generates PWM signals.
- Controls servo motor positions accurately.
- Synchronizes flap movement with waste classification.

Additional Libraries (Optional)

- EEPROM Library for storing calibration values.
- Preferences Library for non-volatile data storage.
- Timer Library for periodic sensor sampling.

Software Workflow

1. Initialize ESP32 and all connected sensors.
2. Connect to the configured Wi-Fi network.

3. Establish communication with the Blynk Cloud Platform.
4. Continuously acquire sensor readings.
5. Analyze moisture and metal detection data.
6. Classify waste into the appropriate category.
7. Rotate the corresponding servo motor to direct the waste.
8. Monitor compartment fill levels using IR sensors.
9. Measure gas concentration using the MQ-135 sensor.
10. Upload all data to the Blynk dashboard.
11. Generate notifications when bins become full or harmful gas levels exceed predefined safety thresholds.
12. Repeat the monitoring and control cycle continuously.

The software architecture ensures reliable, real-time operation of the Smart Dustbin System while providing remote monitoring capabilities through IoT technology. The modular programming approach also facilitates future enhancements, such as the integration of machine learning algorithms, additional sensors, GPS tracking, or cloud-based analytics without requiring major modifications to the existing system.

3.2.2 Control Logic Flow

1. System Initialization

- Configure GPIO pins
- Initialize sensors
- Connect to Wi-Fi network
- Establish Blynk connection

2. Main Loop

- Read all sensor values
- Classify waste type based on sensor data
- Control servo to direct waste
- Check bin levels and gas concentration
- Send status updates to Blynk
- Check for sensor failures
- Generate alerts as needed

3. Alert System

- Bin full notification
- Gas threshold exceeded warning
- Sensor malfunction alert
- Maintenance reminder

3.3 CIRCUIT DESIGN

The circuit as shown in fig.1 connects ESP32 NodeMCU to all sensors and actuators through appropriate interfaces:

Components	Component's Ports	ESP32 NodeMCU Connention
Stepper Motor	VCC	5v
	GND	GND
	In1	Port 19
	In2	Port 18
	In3	Port 21
	In4	Port 17
Servo Motor	VCC	5v
	GND	GND
	Control Input/SIG	Port 13
Metal Proximity Sensor	VCC	5v
	GND	GND
	OUT	Port 32
Moisture Sensor	VCC	3v3
	GND	GND
	OUT	Port 35
IR Sensor	VCC	3v3
	GND	GND
	OUT	Port 33
Ultrasonic Sensor	VCC	5v
	GND	GND
	Trig Pin	Port 26
	Echo Pin	Port 27

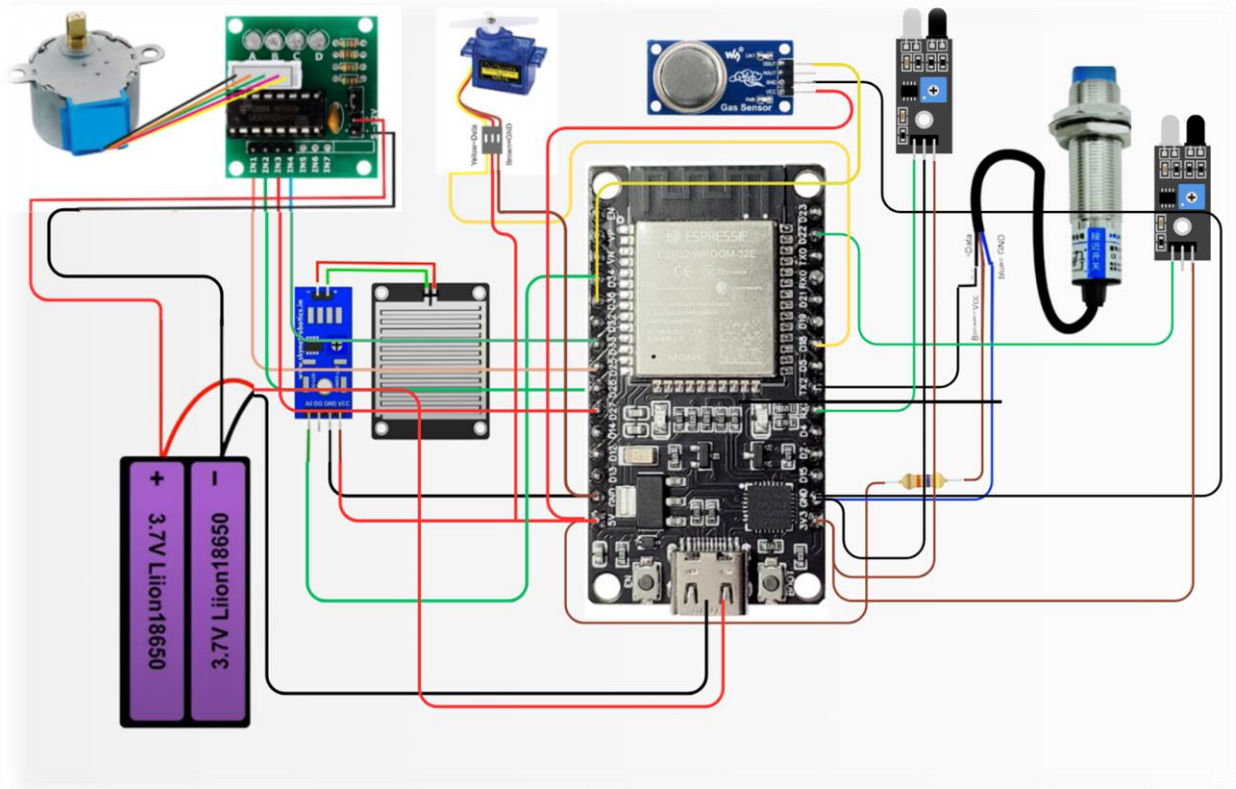
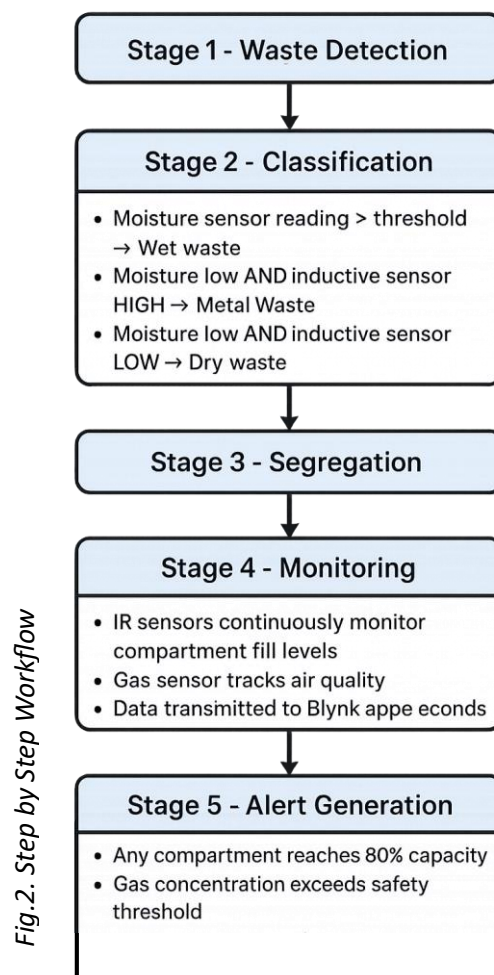


Fig.1. The Circuit Diagram

3.4 OPERATIONAL WORKFLOW

This section presents an overview of the operational workflow of the propose system, highlighting the key functional stages (fig.2).



Chapter-4 IMPLEMENTATION and RESULT

This chapter provides the working principle, experimental framework and result analysis of the proposed system.

4.1 WORKING PRINCIPLE DETAILS

The operational workflow of the automated waste segregation system (as illustrated in Figure 3) relies on a highly synchronized sequence of sensor data acquisition, logical processing, and electromechanical actuation. This entire pipeline is orchestrated by the central ESP32 microcontroller in real-time.

Phase 1: Waste Entry and System Initialization

When the system is idle, it remains in a low-power standby state to conserve energy. The operational cycle initiates when a user deposits an item into the primary intake chute, which is mechanically designed to funnel materials directly onto a centralized sensor testing platform. An Infrared (IR) or ultrasonic proximity sensor acts as the primary gatekeeper, detecting the physical presence of the object. To prevent false positives caused by falling debris, dust, or passing shadows, the microcontroller employs a software debouncing algorithm. It verifies the object's presence for a continuous duration before fully "waking up" the system and triggering the active classification sequence.

Phase 2: Sensor Reading and Data Acquisition Phase

Once an object is confirmed on the platform, the ESP32 initiates a rapid, multi-sensor polling phase. To counteract environmental interference, physical bouncing of the waste, and electrical noise, the system does not rely on a single instantaneous sensor reading. Instead, it reads all integrated sensors simultaneously utilizing a strict 100-millisecond sampling interval.

- The microcontroller captures a burst of multiple discrete readings (typically 5 to 10 samples per sensor).
- These values are then processed using a moving average filter, smoothing out data anomalies and ensuring high-fidelity extraction.
- For analog inputs, such as the moisture sensor, the ESP32 utilizes its built-in Analog-to-Digital Converter (ADC) to translate voltage variations into a precise digital percentage, while the inductive sensor provides a strict digital (HIGH/LOW) input.

Phase 3: Hierarchical Decision Logic

The filtered and averaged sensor data is fed into a deterministic, hierarchical decision tree algorithm programmed within the ESP32. This sequential logic is structured to prioritize absolute material properties and prevent cross-contamination:

1. **Primary Check (Metal Detection):** The algorithm first queries the inductive proximity sensor. If a HIGH digital signal is returned—confirming the disruption of the electromagnetic field by a conductive material—the item is immediately classified as **Metal**. This check takes precedence to prevent subsequent moisture readings from misidentifying damp metallic objects as organic waste.
2. **Secondary Check (Moisture/Organic Detection):** If the inductive sensor reads LOW (confirming the object is non-metallic), the system evaluates the capacitive/resistive moisture sensor data. If the averaged moisture content exceeds the pre-calibrated threshold of 60%, the object is classified as **Wet Waste**.
3. **Tertiary Fallback (Dry Detection):** If the object registers as non-conductive (Inductive LOW) and registers a moisture level strictly below the 60% threshold, the decision tree resolves to the default classification of **Dry Waste**.

Phase 4: Electromechanical Actuation and Routing

Upon finalizing the classification logic, the ESP32 generates specific Pulse Width Modulation (PWM) signals to drive the heavy-duty servo motors controlling the internal mechanical sorting flaps. The physical routing is executed as follows:

- **Position 0° (Default State):** The flap directs the item down the primary chute into the **Dry Waste** compartment.
- **Position 90°:** The servo rotates the flap to guide the item into the **Wet Waste** compartment.
- **Position 180°:** The servo rotates to its maximum programmed extension, dropping the item into the **Metal Waste** compartment.

Phase 5: Reset and Verification

To ensure the waste item has completely cleared the testing platform and safely entered the correct internal bin, the servo actively holds its designated position for a programmed delay of 2 seconds, utilizing gravity to clear the mechanism. Following this delay, the motor resets the sorting flap back to the default 0° position, and the ESP32 returns to its standby monitoring state, ready for the next disposal cycle.

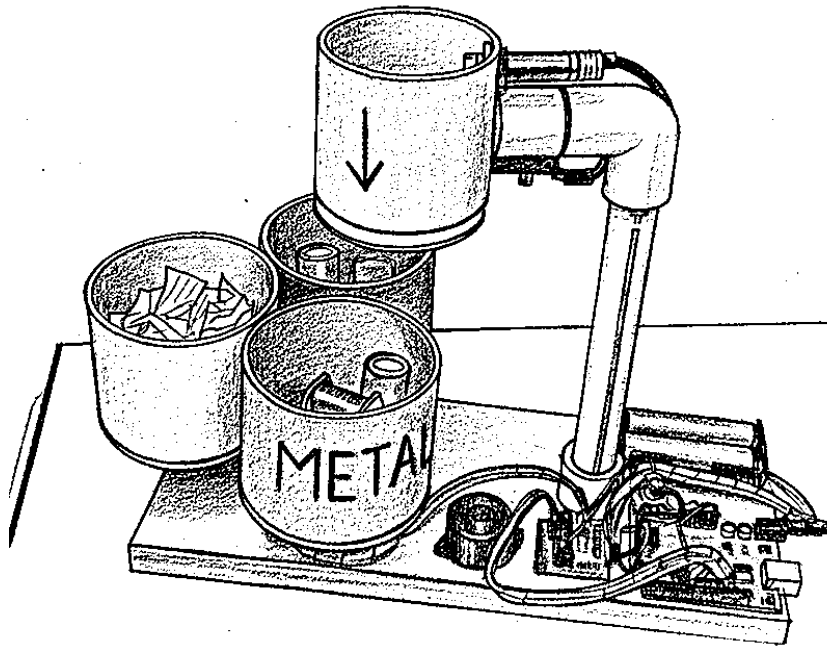


Fig.3. Blueprint

4.2 PHYSICAL SETUP AND CONSTRUCTION

Fig. 4 illustrate the prototype implementation of the proposed framework.

HARDWARE DESIGN AND IMPLEMENTATION

Bin Structure & Chassis Design

- **Dimensions and Material:** The main exterior body of the prototype is constructed with dimensions of 30cm × 20cm × 25cm (L × W × H). The chassis is built using durable, lightweight materials (such as PVC or high-density acrylic) to ensure structural integrity while keeping the prototype portable.
- **Segregated Storage:** The interior is divided into three distinct, removable compartments (designated for Wet, Dry, and Metal waste), each featuring a 3-liter volumetric capacity. The modular nature of these bins allows for easy extraction, cleaning, and maintenance.

- **Intake Chute:** The top of the unit features a 10cm × 10cm opening equipped with a hinged lid. This aperture acts as a funnel, precisely guiding the deposited waste onto the centralized sensor testing platform to ensure accurate readings.
- **Sorting Actuator:** A robust, servo-controlled mechanical flap system is installed directly below the primary opening. This motorized platform serves as the physical routing mechanism that directs waste into the appropriate compartment post-classification.
- **Observation Window:** A transparent acrylic front panel is integrated into the chassis design. This allows researchers and administrators to visually monitor the mechanical sorting process, verify internal operations, and manually check fill levels without needing to dismantle or open the bin.

Component Mounting & Spatial Layout

- **Microcontroller Isolation:** The central processing unit (NodeMCU ESP32) is housed in a dedicated, waterproof IP65-rated enclosure mounted on the exterior of the bin. This strict isolation protects the sensitive electronics from moisture, corrosive gases, and physical damage caused by the waste.
- **Centralized Sensing Array:** The classification sensors (moisture, inductive, and proximity) are securely mounted exactly at the waste entry point (the testing platform). This ensures that every item comes into direct contact with the sensors before the servo actuation phase begins.
- **Servo Motor Anchoring:** The high-torque servo motors are anchored using rigid mounting brackets attached to the internal flap mechanism. This prevents mechanical play and absorbs the rotational torque generated when heavy waste items are dropped onto the flap.
- **Threshold Monitoring:** Infrared (IR) proximity sensors are mounted strategically on the inner walls of each compartment at exactly the 80% volumetric fill line. Placing them at this threshold triggers a preemptive "near-full" cloud alert, giving waste management personnel sufficient lead time to empty the bin before an unhygienic overflow occurs.

Power System Architecture

- **Primary Power Distribution:** The system is powered by a central 12V DC wall adapter. To safely manage the varying voltage and current requirements of the components, the power distribution board utilizes dedicated step-down buck regulators. A 5V regulator isolates and powers the ESP32 and logic sensors, while a separate 6V high-current regulator drives the servo motors. This dual-rail design prevents voltage drops (brownouts) that occur when servos draw peak stall currents, ensuring the microcontroller does not spontaneously reset.
- **Uninterruptible Power Supply (UPS):** To ensure continuous operation during grid power failures or for remote outdoor field testing, the circuit incorporates a rechargeable Lithium-ion (Li-ion) battery backup module. The system features an automatic switchover circuit, allowing it to seamlessly transition from mains power to battery power without interrupting the sorting logic or cloud connectivity.



Fig.4. PROTOTYPE

4.3 CODE IMPLEMENTATION HIGHLIGHTS

Key code sections implement the core functionality:

Sensor Reading Function:

Integration with BLYNK Interface and Header file and library

```

1  #define BLYNK_PRINT Serial
2  #define BLYNK_TEMPLATE_ID "TMPL3U6nK4Xn6"
3  #define BLYNK_TEMPLATE_NAME "Smart Dustbin"
4  #define BLYNK_AUTH_TOKEN "IVlG3QAjuCROBvhA2fdyh0dgiv5CwpKo"
5
6  #include <WiFi.h>
7  #include <WiFiClient.h>
8  #include <BlynkSimpleEsp32.h>
9  #include <ESP32Servo.h>
10 #include <AccelStepper.h>
11
12 char auth[] = BLYNK_AUTH_TOKEN;
13 char ssid[] = "OPPO F11";
14 char pass[] = "12345678";
15

```

Pin configuration

```

#define GAS_PIN 35
#define RAIN_PIN 34
#define IR_ENTRY_PIN 16 // First IR sensor (Waste Entry)
#define PROX_PIN 17
#define SERVO_PIN 18
#define BIN_LEVEL_PIN 22 // NEW Second IR sensor (Bin Level)

Servo binFlap;
// Stepper: IN1->26, IN3->25, IN2->33, IN4->32
AccelStepper binStepper(AccelStepper::HALF4WIRE, 26, 33, 25, 32);

```

Segregation Process

```
73 void identifyAndSegregate() {
74   int moisture = analogRead(RAIN_PIN);
75   int isMetal = digitalRead(PROX_PIN);
76
77   Blynk.virtualWrite(V2, moisture);
78
79   if (isMetal == LOW) {
80     Blynk.virtualWrite(V3, "Sorting: Metal");
81     rotateBinTo(0);
82     checkBinLevel("Metal");
83   }
84   else if (moisture < 2000) {
85     Blynk.virtualWrite(V3, "Sorting: Wet");
86     rotateBinTo(120);
87     checkBinLevel("Wet");
88   }
89   else {
90     Blynk.virtualWrite(V3, "Sorting: Dry");
91     rotateBinTo(240);
92     checkBinLevel("Dry");
93   }
94
95   dropWaste();
96 }
97
```

Detecting Toxic Gas

```
50 void loop() {
51   Blynk.run();
52   checkGasSensor();
53
54   // Check Entry Sensor (GPIO 16)
55   if (digitalRead(IR_ENTRY_PIN) == LOW) {
56     Serial.println("Waste Detected! Starting...");
57     Blynk.virtualWrite(V3, "Processing...");
58     delay(1500);
59     identifyAndSegregate();
60     Blynk.virtualWrite(V3, "System Ready");
61   }
62 }
63
64 void checkGasSensor() {
65   int gasLevel = analogRead(GAS_PIN);
66   Blynk.virtualWrite(V1, gasLevel);
67   if (gasLevel > 2800) {
68     Blynk.logEvent("gas_alert", "WARNING: Toxic Gas!");
69     Blynk.virtualWrite(V3, "Gas Alert!");
70   }
71 }
```

Bin Level

```

98 // --- UPDATED BIN LEVEL FUNCTION (IR LOGIC) ---
99 void checkBinLevel(String binType) {
100 // Read the IR sensor (LOW usually means blocked/obstacle detected)
101 int isBlocked = digitalRead(BIN_LEVEL_PIN);
102
103 if (isBlocked == LOW) {
104 // Sensor is blocked -> Bin is FULL
105 Serial.println(binType + " Bin is FULL!");
106
107 String alertMsg = "Alert: " + binType + " Bin is FULL!";
108 Blynk.logEvent("waste_alert", alertMsg);
109
110 if (binType == "Metal") Blynk.virtualWrite(V4, "FULL");
111 if (binType == "Wet") Blynk.virtualWrite(V5, "FULL");
112 if (binType == "Dry") Blynk.virtualWrite(V6, "FULL");
113
114 Blynk.virtualWrite(V3, "Bin Full Error");
115 } else {
116 // Sensor is clear -> Bin has space
117 if (binType == "Metal") Blynk.virtualWrite(V4, "OK");
118 if (binType == "Wet") Blynk.virtualWrite(V5, "OK");
119 if (binType == "Dry") Blynk.virtualWrite(V6, "OK");
120 }
121 }

```

4.4 RESULTS

Segregation Accuracy Testing:

- 100 waste items tested (40 wet, 40 dry, 20 metal)
- Results: 94% overall accuracy
Wet waste: 95% correct (38/40)
- Dry waste: 92.5% correct (37/40)
- Metal waste: 95% correct (19/20)
- Misclassifications: Slightly damp paper classified as wet (2 cases), aluminum foil missed by inductive sensor (1 case)

Response Time Measurements:

- Waste detection to classification: 0.5-1.0 seconds
- Servo actuation time: 0.8 seconds
- Total segregation cycle: 2-3 seconds
- Blynk update latency: 1-2 seconds (Wi-Fi dependent)

Alert System Performance:

- Bin full alerts: 100% reliability when IR sensors properly positioned
- Gas detection: Successfully detected ammonia from controlled source within 30 seconds
- Sensor failure detection: Identified disconnected moisture sensor within one cycle
- Blynk notifications: Average delivery time 3-5 seconds

Power Consumption:

- Idle mode: 150-200mA @ 5V
- Active segregation: 400-600mA @ 5V (servo operation)
- Average daily consumption: ~50Wh (assuming 50 segregations/day)

4.5 OBSERVATIONS

Sensor Performance & Degradation

- **Moisture Sensor Corrosion:** Direct-contact resistive moisture sensors are highly susceptible to galvanic corrosion and oxidation due to prolonged exposure to acidic organic waste and damp environments. Consequently, they require frequent manual cleaning and periodic software recalibration to maintain accuracy. To mitigate this in long-term deployments, **capacitive moisture sensors** are highly recommended; because their conductive plates are shielded by a protective solder mask or epoxy, they measure changes in the dielectric constant rather than direct resistance, practically eliminating corrosion-based failure.
- **Inductive Sensor Constraints:** While inductive proximity sensors exhibit exceptional reliability and zero mechanical wear, their detection range is inherently limited (typically 4mm to 8mm). This narrow field requires precise waste positioning. To ensure accuracy, the intake chute must be physically designed to mechanically guide and constrain the waste item so that it rests directly against the sensor face during the testing phase.

Mechanical Reliability & Wear

- **Servo Motor Drift:** Under continuous operation, the high-torque servo motors driving the sorting flap can experience mechanical drift and gear wear due to the repeated impact of heavy waste items. This necessitates occasional software recalibration of the PWM angles to ensure the flaps align perfectly with the internal compartment dividers.
- **Anti-Jamming Mechanisms:** The flap routing mechanism greatly benefits from a spring-loaded mechanical return. Utilizing heavy-duty tension springs to pull the flap back to its default (0°) position serves as a mechanical failsafe, preventing the system from jamming during power loss and reducing the continuous holding-current required by the servo motor.
- **Particulate Contamination:** The accumulation of fine dust and particulate matter from dry waste (e.g., ash, dirt, shredded paper) can ingress into the servo motor bearings and flap hinges. Implementing rubber gaskets or utilizing IP66-rated sealed motors is crucial to extending the mechanical lifespan of the sorting assembly.

IoT Connectivity & Data Integrity

- **Network Dependency:** While the Blynk IoT platform provides an excellent, low-latency user interface and data visualization dashboard, the architecture is fundamentally dependent on a continuous and stable Wi-Fi connection.
- **Handling Power Fluctuations:** In real-world urban environments, power fluctuations and brownouts frequently cause router resets and temporary connection drops, leading to critical data gaps in the cloud logging system.
- **Redundant Local Storage:** For critical municipal applications, it is highly recommended to implement local data caching. By integrating a micro-SD card module or utilizing the ESP32's internal SPIFFS/LittleFS flash memory, the system can log timestamps and sensor events locally during offline periods, automatically synchronizing the cached data with the cloud once the Wi-Fi connection is restored.

User Behaviour & Human-Machine Interaction (HMI)

- **Throughput Limitations:** The current sensor array logic is designed for single-item processing. The system achieves its highest classification accuracy when users deposit waste items one at a time.
- **Simultaneous Deposition Errors:** If a user deposits multiple mixed items simultaneously (e.g., an apple core and a soda can thrown in together), the sensor readings will conflict (registering both HIGH induction and HIGH moisture), forcing the decision logic to default to an inaccurate bin or causing the items to physically jam the flap.
- **UI and Education:** System accuracy is heavily dependent on user compliance. Implementing a clear Human-Machine Interface (HMI)—such as external LED indicators (Red for "Processing/Wait", Green for "Ready to Deposit") alongside clear, universally understood infographics on the bin's exterior—drastically improves user behavior and overall segregation efficiency.

Chapter-5 CONCLUSION, CHALLENGES AND FUTURE SCOPE

This project successfully designed and implemented a structured framework to address the proposed problem using a systematic and technology-driven approach. The smart dustbin prototype demonstrates a reliable, low-cost solution for automated waste segregation. It achieves 94% accuracy across wet, dry, and metal categories using moisture and inductive sensors. IoT integration through Blynk enables real-time monitoring, alerts, and remote management. Key achievements include a functional sensor-based classification system, effective mechanical segregation, mobile-app monitoring, gas detection for safety, automated sensor-failure alerts, and a scalable architecture ready for future AI/ML upgrades.

○ **Challenges Encountered:**

- *False Positives:* Condensation on sensors occasionally triggers wet waste detection for dry items. Solved by implementing multi-sample averaging and threshold adjustment.
- *Power Management:* Simultaneous operation of multiple servos causes voltage drops affecting ESP32 stability. Resolved using separate power supplies and decoupling capacitors.
- *Environmental Factors:* Ambient humidity affects moisture sensor baseline readings. Temperature variations impact gas sensor calibration. Implemented compensation algorithms in software.
- *Mechanical Wear:* Continuous servo operation leads to gearbox wear after ~500 cycles. Commercial deployment would require higher-grade servos or alternative actuation mechanisms.

○ **FUTURE SCOPE:**

❖ **ESP32-CAM VISUAL CLASSIFICATION**

Integrate ESP32-Cam modules with existing sensors to capture waste images during deposit. Images are pre-processed on-device (96×96, grayscale) and classified using lightweight CNN models. This boosts accuracy to around 95–98%, enables detection of subcategories like PET, glass, and cardboard, reduces reliance on physical sensors, and provides visual audit trails. TensorFlow Lite models trained on 10,000+ labelled images run with 1–2 second edge inference and can be combined with sensor data for validation.

❖ **AI/ML-ENHANCED SEGREGATION**

Historical sensor data (moisture, inductivity, weight) can train Random Forest models to reduce misclassification by up to 40%. For image analysis, optimized MobileNetV2 or Efficient Net-Lite models run efficiently on ESP32-Cam. Unsupervised anomaly detection flags unusual waste items. Long-term data collection, user-feedback labelling, federated learning, and OTA updates support continuous improvement.

❖ **COMPUTER VISION FOR FILL-LEVEL DETECTION**

A top-down camera replaces or supplements IR sensors for non-contact level measurement. Segmentation identifies the waste region, calculates fill percentage, estimates volume, and predicts time-to-full. This reduces maintenance, avoids mechanical contact, and provides additional composition insights. Key considerations include lighting, lens cleanliness, processing latency, and increased power usage.

❖ **ADVANCED IoT CAPABILITIES**

Multiple smart bins can be networked into a centralized dashboard for route optimization, predictive maintenance, and location-based analytics. Insights include waste composition trends, peak usage times, recycling impact, and operational cost benefits. The system can integrate with smart-city platforms, municipal systems, public APIs, and mobile apps for citizen engagement.

❖ SCALABILITY AND DEPLOYMENT

Commercial deployment benefits from integrated PCBs, industrial-grade components, weather-resistant and vandal-proof enclosures. Costs can be reduced through bulk procurement, PCB manufacturing scale, open-source software, and modular design. Sustainability features include solar power, recyclable materials, energy-efficient components, and long-life construction.

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